

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 2: Polynomials

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Zeroes and coefficients of a quadratic polynomial

1. The zeroes of $p(x) = 2x^2 - k\sqrt{2}x + 1$ add up to $\sqrt{2}$. Find k . (a) $\sqrt{2}$ (b) 2 (c) $2\sqrt{2}$ (d) $\frac{1}{2}$

CBSE 2024, 1 mark

SOLUTION

$$\text{Sum of zeroes} = -\frac{b}{a} = \frac{k\sqrt{2}}{2}$$

$$\frac{k\sqrt{2}}{2} = \sqrt{2} \text{ gives } k = 2$$

Answer: (b)

2. Each zero of $x^2 + px + q$ is double a zero of $4x^2 - 5x - 6$. Find p . (a) $-\frac{5}{2}$ (b) $\frac{5}{2}$ (c) -5 (d) 10

CBSE 2024, 1 mark

SOLUTION

Let the zeroes of $4x^2 - 5x - 6$ be α and β . Then $\alpha + \beta = \frac{5}{4}$.

The zeroes of $x^2 + px + q$ are 2α and 2β , with sum $2(\alpha + \beta) = \frac{5}{2}$.

$$\text{Sum of zeroes} = -p, \text{ so } p = -\frac{5}{2}$$

Answer: (a)

2 Zeroes from the graph

3. Assertion (A): a polynomial whose graph touches the x -axis at exactly one point cannot be quadratic. Reason (R): a polynomial of degree n , with $n > 1$, has at most n zeroes. Choose: (a) A and R are both true and R explains A. (b) A and R are both true but R does not explain A. (c) A is true and R is false. (d) A is false and R is true. CBSE 2024, 1 mark

SOLUTION

A is false: the graph of $(x - 1)^2 = x^2 - 2x + 1$ touches the x -axis only at $x = 1$, and it is a quadratic polynomial.

R is true: a polynomial of degree n has at most n zeroes.

Answer: (d)

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